

Information Technology University of the Punjab
SE460T Cloud Computing with Big Data Applications – Fall 2025
Quiz 3 Section A – December 17, 2025

Name: _____

Roll No.: _____

Time allowed: 25 minutes

Maximum Marks: 22

1. ETL and ELT are related to all stages of a data pipeline except which one? [1]

Analysis and Visualization

2. What are the 6 steps in data wrangling? Just provide the terms for the steps without any details. [3]

Discover, Clean, Structure, Enrich, Validate, Publish

3. From which type of data sources is AWS Data Exchange used to ingest? [1]

Third-party datasets that are uploaded on AWS.

4. Provide the name of an AWS service that can be considered a data warehouse. [1]

Amazon Redshift

5. An ELT pipeline typically loads the data in a data lake and an ETL pipeline loads in a data warehouse. [1]

6. In the Enriching phase of data wrangling, give an example that illustrates how data from various sources is enriched. [2]

Two datasets are merged and supplemented by adding ticket_type and sales region column.

Combine rows from supp1 with rows from supp2.					Add a ticket_type category for supp2 rows.		Add the sales region.		
A	B	C	D	E		F	G	H	
ticket_id	customer_id	subject	status	priority	ticket_type	create_date	updated_date	solved_date	sales_region
900865	9999999	Error message 808	Closed	Medium	Problem	2020-06-26 02:45:29	2020-06-27 21:31:48	2020-06-27 23:30:09	
900866	2020411	Nullpointer exception on save	Open	High	Problem	2020-04-02 12:15:00	2020-04-05 11:15:00		USEast
900867	2022010	Question about grades report	Open	High	Question	2020-04-06 10:56:00			USWest
900862	1744899	Nightly batch failing	Closed	Medium	Problem	2020-01-04 08:06:00	2020-04-31 09:00:00	2020-04-31 09:00:00	Europe
900865	1744005	Intermittent error message	Open	Low	Problem	2020-01-06 11:14:00			Europe

7. What activities are performed during the data validation phase of data wrangling? [2]

Audit your work:

Count expected rows. Check consistency. Check expected formats and data types. Check for duplicates. Check for PII. Check for outliers.

Fix the data:

Address audit findings. Address issues at the source. Fix issues with integration tools or scripts.

Note: Audit and fix must be mentioned in student's answer. Providing some detail and not all in each section of Audit and Fix is okay.

8. A json is imported into Excel which parses the json document. Which step of data wrangling can this be considered? [1]

Data structuring

9. What are AWS Glue workers, and which processing engine do they use? [2]

AWS Glue workers are logical entities of computation in AWS Glue. They use Apache Spark engine as the underlying distributed processing region.

10. What does the AWS DataSync service do? At what stage or stages of the data pipeline is it used? [2]

AWS DataSync is a managed service that automates and accelerates the secure transfer of large amounts of data between on-premises storage and AWS services.

It is used mainly at the data ingestion stage of the data pipeline.

11. What is data cataloging? Which AWS Glue feature automatically performs cataloging on a given dataset? Where is the data catalog stored once Glue creates it? [3]

Data cataloging is the process of discovering, classifying, and storing metadata about datasets.

Glue Crawler

In a database created inside Glue Catalog.

12. Which stage or stages of the data pipeline do AWS Glue Jobs belong to? [1]

Transformation (Processing)

13. How can you ensure that one job in an AWS Glue Workflow runs only when another has completed its run?

By adding a trigger at the end of Job A and attaching the trigger to Job B. In the trigger, the option selected should be of “event”.